

object_detection_test

June 23, 2025

```
[3]: import sys
import os
sys.path.append(os.path.join(os.path.dirname(os.getcwd())))
```

```
[1]: # Use deployed HF model for inference

from huggingface_hub import hf_hub_download
from ultralytics import YOLO

# Download the model from the Hugging Face Hub
model_path = hf_hub_download(
    repo_id="lmarte17/gridiron-ai-football-detector",
    filename="best.pt",
    cache_dir="../models"
)
```

best.pt: 0% | 0.00/52.0M [00:00<?, ?B/s]

```
[2]: # Load the model
model = YOLO(model_path)

# Run inference on train image
results = model('../data/original/train/images/
↳57558_002886_Sideline_frame204_jpg.rf.386300f508e1beea9a6034e14cb6fca8.jpg',
↳conf=0.25)
results[0].show()
```

image 1/1 /Users/lmarte/Documents/Projects/CU-Boulder/Deep_Learning/gridiron-ai/notebooks/./data/original/train/images/57558_002886_Sideline_frame204_jpg.rf.386300f508e1beea9a6034e14cb6fca8.jpg: 640x640 11 defenses, 5 olines, 1 qb, 1 ref, 1 running_back, 4 wide_receivers, 139.9ms
Speed: 1.8ms preprocess, 139.9ms inference, 4.5ms postprocess per image at shape (1, 3, 640, 640)

```
[3]: # Test on screenshot
results = model('youtube_screenshots/pass_0001_88.jpg', conf=0.25)
results[0].show()
```

image 1/1 /Users/lmarte/Documents/Projects/CU-Boulder/Deep_Learning/gridiron-ai/notebooks/youtube_screenshots/pass_0001_88.jpg: 384x640 11 defenses, 5 olines, 1 qb, 1 ref, 2 running_backs, 5 wide_receivers, 125.0ms
Speed: 2.0ms preprocess, 125.0ms inference, 0.9ms postprocess per image at shape (1, 3, 384, 640)

```
[5]: boxes = results[0].boxes
    if boxes is not None:
        print(f"Detected {len(boxes)} objects")
        for i, box in enumerate(boxes):
            conf = box.conf[0]
            cls = int(box.cls[0])
            print(f"  Object {i+1}: Class {cls}, Confidence: {conf:.2f}")
```

Detected 25 objects

- Object 1: Class 4, Confidence: 0.84
- Object 2: Class 0, Confidence: 0.84
- Object 3: Class 0, Confidence: 0.83
- Object 4: Class 1, Confidence: 0.81
- Object 5: Class 0, Confidence: 0.80
- Object 6: Class 0, Confidence: 0.79
- Object 7: Class 6, Confidence: 0.78
- Object 8: Class 0, Confidence: 0.76
- Object 9: Class 6, Confidence: 0.75
- Object 10: Class 6, Confidence: 0.73
- Object 11: Class 0, Confidence: 0.73
- Object 12: Class 0, Confidence: 0.73
- Object 13: Class 0, Confidence: 0.71
- Object 14: Class 1, Confidence: 0.63
- Object 15: Class 0, Confidence: 0.61
- Object 16: Class 1, Confidence: 0.60
- Object 17: Class 0, Confidence: 0.59
- Object 18: Class 6, Confidence: 0.56
- Object 19: Class 0, Confidence: 0.53
- Object 20: Class 2, Confidence: 0.47
- Object 21: Class 3, Confidence: 0.46
- Object 22: Class 1, Confidence: 0.42
- Object 23: Class 1, Confidence: 0.40
- Object 24: Class 4, Confidence: 0.32
- Object 25: Class 6, Confidence: 0.28